

CONNECT THE DOTS: NAVIGATING NETWORKS

Session 1: Introduction to Graph Theory, BFS, and DFS Algorithms

Course: Computer Science Fundamentals

Topic: Graph Traversal

Target Level: Beginner / Intermediate

1. Introduction: Small World Phenomena & Social Networks

Have you ever wondered how you are connected to a famous celebrity, a world leader, or someone living on another continent through just a few mutual friends? This fascinating concept is known as the **"Six Degrees of Separation"** or the **Small World Phenomenon**.

We live in a massively interconnected world. Every person you know is connected to other people, who are in turn connected to many others. In computer science, we model these real-world interconnections using a mathematical structure called a **Graph**.

2. What is a Graph? Core Concepts

A **Graph** is a collection of points (called *Nodes* or *Vertices*) connected by lines (called *Edges*). Graphs allow computer scientists to model relationships and solve complex navigation, routing, and networking problems.

Nodes (Vertices)

Nodes represent individual entities, points, or objects in a network.

- **Delhi Metro:** Stations (e.g., Rajiv Chowk, Hauz Khas)
- **Social Media:** User Profiles
- **Family Tree:** Individual Family Members

Edges (Connections)

Edges represent the relationships, links, or pathways connecting two nodes.

- **Delhi Metro:** Train tracks connecting stations
- **Social Media:** Friendships or Following links
- **Family Tree:** Parent-Child relationships

Real-World Example: The Delhi Metro Graph

Imagine traveling on the Delhi Metro network. Key interchange points like **Rajiv Chowk** and **Hauz Khas** are **NODES**. The physical railway tracks running between these stations are **EDGES**. Every time you board a train and travel to the next station, you are navigating along an edge from one node to another!

3. Graph Traversal Algorithms: Meet Buddy BFS & Daring DFS

When a computer searches for an item, a path, or a person inside a graph, it uses a systematic process called **Graph Traversal**. The two most fundamental graph traversal algorithms are **Breadth-First Search (BFS)** and **Depth-First Search (DFS)**.

● Buddy BFS (Breadth-First Search)

Motto: "I like to check all nearby friends first before going farther!"

Buddy BFS explores a network level by level, spreading outward in concentric circles like ripples in a pond. First, Buddy visits all immediate neighbors (1 step away). Next, Buddy visits neighbors of those neighbors (2 steps away), and so forth.

- **Data Structure Used:** Queue (First-In, First-Out / FIFO)
- **Key Feature:** Guaranteed to find the *shortest path* in unweighted networks.
- **Best For:** Finding the closest destination or finding shortest routes.

● Daring DFS (Depth-First Search)

Motto: "I love to explore one path all the way to the end before trying another!"

Daring DFS is an adventurous explorer. She picks one direction and goes as deep into the network as possible until reaching a dead end. Once stuck, she **backtracks** to the last intersection and tries a new unexplored branch.

- **Data Structure Used:** Stack (Last-In, First-Out / LIFO) or Recursion
- **Key Feature:** Explores deep branches fully before returning.
- **Best For:** Solving mazes, topological sorting, and exhaustive exploration.

4. Comparison Summary: BFS vs. DFS

Feature	BFS (Buddy BFS)	DFS (Daring DFS)
Search Order	Level-by-level (Ripples in a pond)	Branch-by-branch (Deep dive)
Underlying Data Structure	Queue (FIFO)	Stack (LIFO) / Recursion
Shortest Path Finding	Optimal for unweighted graphs	Not guaranteed to find shortest path
Memory Requirement	Higher (stores entire frontier levels)	Lower (stores current search path)
Primary Application	GPS Navigation, Finding nearest friends	Solving Sudoku/Mazes, Cycle detection

💡 Case Study: The Metro Rescue Scenario

Suppose you accidentally lost your backpack at a metro station.

- If you use **BFS**, you search all stations 1 stop away, then 2 stops away. If your bag is nearby, BFS finds it immediately in the shortest possible time!
- If you use **DFS**, you pick one metro line and follow it all the way to the final terminus station before checking another line. If your bag was actually just 1 stop away on a different line, DFS will take a very long time to find it!

STUDENT WORKSHEET & PRACTICE ASSESSMENT

Session 1: Network Navigation with BFS and DFS

Instructions: Answer the following practice questions based on the concepts covered in this lesson. Write your complete steps, explanations, or diagrams in the designated spaces.

Question 1: Terminology Identification

BASIC

Consider a social media platform like Instagram or LinkedIn.

- What real-world components represent the **Nodes** in this graph?
- What real-world interactions or relationships represent the **Edges**?

Write your answer here...

Question 2: BFS Traversal Execution

INTERMEDIATE

Suppose we have an undirected graph with the following connections:

- Node **A** is connected to Nodes **B** and **C**.
- Node **B** is connected to Nodes **D** and **E**.
- Node **C** is connected to Node **F**.

Starting at Node **A**, list the exact sequence of nodes visited using **Breadth-First Search (BFS)**. (Assume alphabetical tie-breaking for neighbors).

Sequence of visited nodes: _____
Explanation of steps:

Question 3: DFS Traversal Execution

INTERMEDIATE

Using the exact same graph described in Question 2 (Node A connected to B, C; Node B connected to D, E; Node C connected to F):

Starting at Node **A**, list the exact sequence of nodes visited using **Depth-First Search (DFS)**. (Assume alphabetical tie-breaking for neighbors).

Sequence of visited nodes: _____
Explanation of steps:

Question 4: Algorithmic Selection & Real-World Application

ADVANCED

GPS navigation apps like Google Maps or Apple Maps need to find the quickest driving route from your location to a hospital during an emergency.

- a) Which algorithm—BFS or DFS—is better suited for this emergency navigation task?
- b) Provide two technical reasons justifying your choice based on how the selected algorithm operates.

a) Selected Algorithm: _____

b) Reason 1:

Reason 2:

Question 5: Conceptual Analysis & Backtracking

ADVANCED

In Depth-First Search (DFS), what is the term used when the algorithm hits a node with no unvisited neighbors and must move backward to try a different branch? Describe how a **Stack** structure enables this mechanism.

Term: _____

Explanation: